

Forest Sensor Sprint

Find the smartest place to install a future campus tree sensor.

Win condition: Best evidence-based sensor recommendation with accurate measurements and efficient route planning.

THE SCIENCE YOU ARE USING

Microclimate varies in meters. Temperature, humidity, light, and soil moisture can change sharply over short distances. A spot under a dense canopy can be several degrees cooler and far more humid than open lawn a few steps away. Good sensor placement starts with measuring, not guessing.

Evidence-based siting. A useful sensor sits where its readings represent the area you care about and where it will not be disturbed. Teams collect data at field stations, then argue for one location using their own numbers plus an efficient route between checkpoints.

HOW TO BUILD AND RUN IT

- 01 **Calibrate together.** Take one reading as a whole group at a reference spot so every team's tools agree before they split up.
- 02 **Run the field stations.** Rotate through stations measuring temperature, humidity, light, and soil moisture. Record units every time.
- 03 **Estimate tree height.** Stand at the marked distance from the tree, read the angle to the top with the paper clinometer, and estimate canopy height for one station.
- 04 **Plan an efficient route.** Order your checkpoints to minimize backtracking. Note the route on the card.
- 05 **Recommend one site.** Pick one sensor location and defend it with your measurements, not opinion.

Safety: Buddy system outdoors. Stay on approved route. Sun, hydration, and weather check before launch. Allergy: Pollen sensitivity note. Students do not need to touch plants. Wash or wipe hands after pushing the soil probe into the ground.

MATERIALS

Clipboards	4 (one per team)
Paper clinometers	4 (one per team)
Temperature and humidity meters	4 (one per team)
Soil meters	4 (one per team); the soil meter also reads light
Route cards	shared
Pencils	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

First site measured: _____

Second site measured: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

What is the next spot you would check, and why? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	100